

EcoFlow Open Modbus Protocol V1.0			
Last Update on 10th June 2026			
Rule of Constraint			
How to enable Modbus 1. Cable Connection: a. Direct Connection: need to ensure that inverter is connected directly to the PC without any routing intermediary: (Default IP 192.168.1.1 or 192.168.X.1 Incrementing from 1) b. Connecting via a router: • The inverter will automatically be assigned a corresponding sub-IP address; • you can find the specific address details in the Pro APP or on the router. 2. In the EcoFlow Pro APP, change the control mode for the relevant inverter to Modbus control 3. Establish a connection with inverter using the corresponding IP address in the Modbus tool			
Interaction related: a. Communication: Modbus/tcp only b. Data storage method: big end Compatible products: 1. PowerOcean series: PowerOcean Single-phase; PowerOcean Three-phase; PowerOcean Plus Three-phase 2. OCEAN2 series: OCEAN2 Three-phase; OCEAN2 Pulse Single-phase			
Important Notes: 1) Please note that using write register may affect inverter internal scheduling (not recommended for non-professional users) 2) The heartbeat signals (0x025F) need be sent to inverter every 60 seconds. Otherwise, current running function will be stopped 3) Open Modbus Protocol/Tcp currently does not support cascading scenarios. 4) The inverter operates in standard EcoFlow self-consumption mode by default when first enter Modbus mode. 5) All other parameters, such as the feed-in power limit, will be inherited from the app settings the first time Modbus mode is enabled. However, parameters in EcoFlow mode will not be synchronised with those in Modbus mode thereafter, vice versa. 6) Safety protection measures and grid-related safety logic is operating normally even in Modbus control mode.			
Master Request Frame			
MBAP	Transaction ID	2 bytes	During a communication process, the transaction identifiers of the message frames of the master station and the slave station are consistent.
	Protocol ID	2 bytes	Always 0x0000
	Length	2 bytes	Number of bytes after this field (Unit ID + Function Code + Function Data)
	Unit ID	1 byte	Slave ID, Device address
PDU	Function Code	1 byte	0x03 Read holding registers

	Starting Address	2 bytes	The starting register address for reading.
	Number of Registers	2 bytes	Defines the read length (count) beginning at the start address.

Slave response frame

MBAP	Transaction ID	2 bytes	During a communication process, the transaction identifiers of the message frames of the master station and the slave station are consistent.
	Protocol ID	2 bytes	Always 0x0000
	Length	2 bytes	Number of bytes after this field (Unit ID + Function Code + Function Data)
	Unit ID	1 byte	Slave ID, Device address
PDU	Function Code	1 byte	0x03 Read holding registers
	Data length	2 bytes	Number of bytes after this field (Data)
	Data	n=bytes number	The byte length differs by storage type (e.g., coil, register) in Modbus.

Function Code Explanation	Meaning	Amount of Operation
0x03	Read multiple registers	Single or multiple
0x06	Write in single register	Single
0x10	Write in multiple register	Multiple

Notes

1	If the data of a UINT32 or Float field is 0x12345678, the Modbus format should be 56781234
2	If the data of a UINT16 field is 0x1234, the Modbus format should be 1234

Revision History

The revision history tracks the changes made with each update to the document. The current logic and descriptions of the listed registers will not be changed without notice in next version. The latest version of the document includes all updates from previous versions.

Version 1.0 (10 June 2026)

- Trial version.

Region	Parameter Name	Data Type	Unit	Permission	Register Address	Description
Starting address: 40001 (Decimal) onwards						
Product Information Area	Protocol Version	UINT16 (2 bytes)	/	R	0x0000	The four bytes from high to low represent the corresponding version number, e.g., 0x01234567 indicates V1.23.45.67.
Product Information Area	Product Category	UINT16 (2 bytes)	/	R	0x0001	Product category enumeration is as follows: 1. Three-phase 2. Single-phase
Product Information Area	Product Number	UINT16 (2 bytes)	/	R	0x0002	Product number enumeration is as follows: 1: PowerOcean 2: PowerOcean Single Phase 3: PowerOcean Plus
Product Information Area	Device Serial Number (SN)	STRING	/	R	0x0003	16 bytes Represented by 16-bit ASCII code from low to high
Product Information Area	Firmware Version of the Package	UINT32 (4 bytes)	/	R	0x000B	The four bytes from high to low represent the corresponding version number, e.g., 0x0627490C indicates V6.39.73.12.
Product Information Area	Device Communication Address	UINT16 (2 bytes)	/	R	0x000D	Range: 1-254 Default: 1 (0x01)
Starting from 0x0200						
System Basic Capability Area	Grid Connection Code	UINT16 (2 bytes)	/	R	0x0200	Corresponding to national grid standard codes.
System Basic Capability Area	UTC Time	UINT32 (4 bytes)	/	R	0x0201	Note compatibility.
	Time Zone	UINT16 (2 bytes)	/	R	0x0205	Actual time zone multiplied by 4. East Zone 1 is 4, East Zone 2 is 8. West Zone 1 is -4, West Zone 2 is -8. For example, -4 is read as 00FC. However, this time zone is currently unused.
System Basic Capability Area	Current Total Load Power	FLOAT (4 bytes)	W	R	0x0206	
System Basic Capability Area	Current Grid Power	FLOAT (4 bytes)	W	R	0x0208	Positive number: Power draw from grid Negative number: Power Feed to grid
System Basic Capability Area	Current Total Solar Power	FLOAT (4 bytes)	W	R	0x020A	
System Basic Capability Area	Current Total Battery Power	FLOAT (4 bytes)	W	R	0x020C	Positive number: Charging Negative number: Discharging
System Basic Capability Area	System State of Charge (SOC)	UINT16 (2 bytes)	%	R	0x020E	Range: [0-100]
System Basic Capability Area	System Rated Power	UINT32 (4 bytes)	W	R	0x020F	

System Basic Capability Area	System Status	UINT32 (4 bytes)	/	R	0x0211	Bit field representation: BIT0: Grid-connected/off-grid status, 0: Grid-connected, 1: Off-grid. BIT1: System status, 0: Normal, 1: Abnormal. BIT2: System operational status, 1: On work, 0: Standby BIT3: Low power enable status BIT4~6: Operating status detail (enumeration), 0: Standby mode, 1: Self-consumption mode, 2: AI mode (Coming Soon) BIT7-10: Control Mode 0: Default Mode (No special functions; functionality follows system mode) 1: System Power Feed/Draw (based on electricity meter) 2: Inverter Power Feed/Draw (based on inverter) 3: Battery Charge/Discharge (based on battery) BIT11: Manual Mode Status 0: Not active 1: Active BIT12: BMS Connection Status 0: Not connected 1: Connected
System Basic Capability Area	System State 2	UINT32 (4 bytes)	/	R	0x0213	Bit field representation: BIT0: System Shutdown Enable BIT1: Upgrade Process Shutdown BIT2: EPO Triggered BIT3: Low Power Mode Triggered BIT4: Fan Failure BIT5: System Failure BIT6: BP Reverse Connection BIT7: BP Disconnection BIT8: Auxiliary Power Failure BIT9: PCS Timeout BIT10: PCS Failure BIT11: IGBT Self-Test Failure BIT12: NTC High-Temperature Protection BIT13: BP Overheating BIT14: NTC Circuit Failure BIT15: System Reset BIT16: Hardware Version Error BIT17: Parallel Inverter Power Synchronisation Error BIT18: NTC Low-Temperature Protection BIT19: Parallel Master-Slave Inverter Conflict BIT20: Parallel Slave Inverter setting Error BIT21: Parallel Inverter Error BIT22: Parallel Meter Fault
System Basic Capability Area	System Control Command	UINT32 (4 bytes)	/	W	0x0215	Bitfield representation, where 1 indicates enabled: BIT0: Active off-grid BIT1: System shutdown BIT3: Power-saving mode switch BIT4-7: Control method. 0: Default mode (no special functions, functions are based on system mode) 1: System Power Draw/Feed (based on electricity meter) 2: Inverter Power Draw/Feed (based on inverter) 3: Battery Power Charge/Discharge (limit values, based on battery)
System Basic Capability Area	Backup Power Ratio	UINT16 (2 bytes)	%	RW	0x0217	Range: [0-100]

System Basic Capability Area	Grid Feed Mode Setting	UINT16 (2 bytes)	/	R	0x0218	Grid feed mode setting: 0: Limited mode 1: Unlimited mode 2. Limited mode (percentage method)
System Basic Capability Area	Maximum Grid Feed Power	UINT32 (4 bytes)	W	R	0x0219	Valid when grid feed mode is set to limited mode.
System Basic Capability Area	Circuit Breaker Capacity Setting	UINT16 (2 bytes)	A	R	0x021B	Ocean 2/PowerOcean Plus: Range: 5-400A. Default: 400A. PowerOcean 3ph: Range: 16-400A. Default: 32A. PowerOcean single-phase: Range: 0-120A. Default: 120A.
System Basic Capability Area	Device Indicator Brightness	UINT16 (2 bytes)	%	RW	0x021C	Range: 0-100.
System Basic Capability Area	System Power Draw/Feed Setting	INT32 (4 bytes)	W	RW	0x021D	Positive number: Device draws power from the grid. Negative number: Device feeds power to the grid. Activation conditions: 1. Control method is system power feed/draw . 2. Meter is functioning normally. 3. Other limiting factors: a. Grid power draw limit: If the current device load consumption and battery charging capacity exceed the set power, the maximum draw can be achieved; otherwise, the actual draw will be lower. b. Grid power feed limit: The remaining capacity after PV and battery supply to the load must exceed the absolute value of the set feed power.
System Basic Capability Area	Inverter Power Draw/Feed Setting	INT32 (4 bytes)	W	RW	0x021F	Positive number: Device draws power from the grid. Negative number: Device feeds power to the grid. Activation conditions: 1. Control method is inverter power feed/draw . 2. Other limiting factors: a. Inverter power draw limit: If the current device load consumption and battery charging capacity exceed the set power draw, the maximum draw can be achieved; otherwise, the actual draw will be lower. b. Inverter power feed limit: The remaining capacity after PV and battery supply to the load must exceed the absolute value of the set feed power.
System Basic Capability Area	Inverter Maximum Inverter Capacity	UINT32 (4 bytes)	W	R	0x0221	
System Basic Capability Area	Inverter Maximum Rectifier Capacity	UINT32 (4 bytes)	W	R	0x0223	
System Basic Capability Area	Inverter Output Power	INT32 (4 bytes)	W	R	0x0225	Rectification positive, inversion negative.
System Basic Capability Area	Battery Capacity	UINT32 (4 bytes)	WH	R	0x0227	
System Basic Capability Area	Battery Discharge Capacity	UINT32 (4 bytes)	W	R	0x0229	
System Basic Capability Area	Battery Charge Capacity	UINT32 (4 bytes)	W	R	0x022B	
System Basic Capability Area	System Working Mode Setting	UINT16 (2 bytes)	/	RW	0x022D	Operating mode setting: 1: Self-consumption mode, 2: AI mode (reserved).
System Basic Capability Area	Grid Phase A Current	FLOAT (4 bytes)	A	R	0x022E	
System Basic Capability Area	Grid Phase B Current	FLOAT (4 bytes)	A	R	0x0230	
System Basic Capability Area	Grid Phase C Current	FLOAT (4 bytes)	A	R	0x0232	
System Basic Capability Area	Phase A Voltage	FLOAT (4 bytes)	V	R	0x0234	
System Basic Capability Area	Phase B Voltage	FLOAT (4 bytes)	V	R	0x0236	
System Basic Capability Area	Phase C Voltage	FLOAT (4 bytes)	V	R	0x0238	

System Basic Capability Area	Battery Charge/Discharge Setting	INT32 (4 bytes)	W	RW	0x023A	Positive number: Device draws power from the grid. Negative number: Device feeds power to the grid. Activation conditions: 1. Control method is battery power feed/draw . 2. Other limiting factors: a. battery power draw limit: If the current device load consumption and battery charging capacity exceed the set power draw, the maximum draw can be achieved; otherwise, the actual draw will be lower. b. battery power feed limit: The remaining capacity after PV and battery supply to the load must exceed the absolute value of the set feed power.
System Basic Capability Area	Limitation Percentage of Feed to Grid	UINT16 (2 bytes)	%	R	0x023C	[1-100] Setting percentage based on rated power
System Basic Capability Area	Battery Voltage	FLOAT (4 bytes)	V	R	0x023D	
System Basic Capability Area	Battery Current	FLOAT (4 bytes)	A	R	0x023F	Positive number: Charging Negative number: Discharging
System Basic Capability Area	Battery Temperature	FLOAT (4 bytes)	°C	R	0x0241	Average Battery Pack temperature
System Basic Capability Area	Inverter Phase A Voltage	FLOAT (4 bytes)	V	R	0x0243	
System Basic Capability Area	Inverter Phase B Voltage	FLOAT (4 bytes)	V	R	0x0245	
System Basic Capability Area	Inverter Phase C Voltage	FLOAT (4 bytes)	V	R	0x0247	
System Basic Capability Area	Inverter Phase A Current	FLOAT (4 bytes)	A	R	0x0249	
System Basic Capability Area	Inverter Phase B Current	FLOAT (4 bytes)	A	R	0x024B	
System Basic Capability Area	Inverter Phase C Current	FLOAT (4 bytes)	A	R	0x024D	
System Basic Capability Area	Inverter Temperature	FLOAT (4 bytes)	°C	R	0x024F	
System Basic Capability Area	Inverter Frequency	FLOAT (4 bytes)	Hz	R	0x0251	
System Basic Capability Area	PV1 Voltage	FLOAT (4 bytes)	V	R	0x0253	
System Basic Capability Area	PV2 Voltage	FLOAT (4 bytes)	V	R	0x0255	
System Basic Capability Area	PV3 Voltage	FLOAT (4 bytes)	V	R	0x0257	
System Basic Capability Area	PV1 Current	FLOAT (4 bytes)	A	R	0x0259	
System Basic Capability Area	PV2 Current	FLOAT (4 bytes)	A	R	0x025B	
System Basic Capability Area	PV3 Current	FLOAT (4 bytes)	A	R	0x025D	
System Basic Capability Area	Heartbeat Signal	UINT16 (2 bytes)	/	W	0x025F	This register must be set to 1 at least once within 60 seconds
System Basic Capability Area	Maximum Grid Feed Power Current Value	UINT32 (4 bytes)	W	R	0x0260	The maximum effective grid feed power limit value of the system after being subject to internal safety regulations.
Starting from 0x0800						
System Basic Capability Area	Current Number of System Faults	UINT16 (2 bytes)	/	R	0x0800	Range: [0-20]. A maximum of 20 faults will be displayed; more than 20 faults will only show the first 20.
System Basic Capability Area	Fault Event 1	UINT16 (2 bytes)	/	R	0x0801	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 2	UINT16 (2 bytes)	/	R	0x0802	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 3	UINT16 (2 bytes)	/	R	0x0803	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 4	UINT16 (2 bytes)	/	R	0x0804	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 5	UINT16 (2 bytes)	/	R	0x0805	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 6	UINT16 (2 bytes)	/	R	0x0806	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 7	UINT16 (2 bytes)	/	R	0x0807	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 8	UINT16 (2 bytes)	/	R	0x0808	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 9	UINT16 (2 bytes)	/	R	0x0809	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 10	UINT16 (2 bytes)	/	R	0x080A	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 11	UINT16 (2 bytes)	/	R	0x080B	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 12	UINT16 (2 bytes)	/	R	0x080C	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 13	UINT16 (2 bytes)	/	R	0x080D	Fault code, refer to fault definition.

System Basic Capability Area	Fault Event 14	UINT16 (2 bytes)	/	R	0x080E	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 15	UINT16 (2 bytes)	/	R	0x080F	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 16	UINT16 (2 bytes)	/	R	0x0810	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 17	UINT16 (2 bytes)	/	R	0x0811	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 18	UINT16 (2 bytes)	/	R	0x0812	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 19	UINT16 (2 bytes)	/	R	0x0813	Fault code, refer to fault definition.
System Basic Capability Area	Fault Event 20	UINT16 (2 bytes)	/	R	0x0814	Fault code, refer to fault definition.
Starting from 0x0820						
System Basic Capability Area	Number of Online Batteries	UINT16 (2 bytes)	/	R	0x0820	Range: 0~12.
System Basic Capability Area	Battery 1 SOC	UINT16 (2 bytes)	%	R	0x0821	Range: 0~100.
System Basic Capability Area	Battery 2 SOC	UINT16 (2 bytes)	%	R	0x0822	Range: 0~100.
System Basic Capability Area	Battery 3 SOC	UINT16 (2 bytes)	%	R	0x0823	Range: 0~100.
System Basic Capability Area	Battery 4 SOC	UINT16 (2 bytes)	%	R	0x0824	Range: 0~100.
System Basic Capability Area	Battery 5 SOC	UINT16 (2 bytes)	%	R	0x0825	Range: 0~100.
System Basic Capability Area	Battery 6 SOC	UINT16 (2 bytes)	%	R	0x0826	Range: 0~100.
System Basic Capability Area	Battery 7 SOC	UINT16 (2 bytes)	%	R	0x0827	Range: 0~100.
System Basic Capability Area	Battery 8 SOC	UINT16 (2 bytes)	%	R	0x0828	Range: 0~100.
System Basic Capability Area	Battery 9 SOC	UINT16 (2 bytes)	%	R	0x0829	Range: 0~100.
System Basic Capability Area	Battery 10 SOC	UINT16 (2 bytes)	%	R	0x082A	Range: 0~100.
System Basic Capability Area	Battery 11 SOC	UINT16 (2 bytes)	%	R	0x082B	Range: 0~100.
System Basic Capability Area	Battery 12 SOC	UINT16 (2 bytes)	%	R	0x082C	Range: 0~100.
Starting from 0x0840						
System Basic Capability Area	Total Metered Energy (Today)	FLOAT (4 bytes)	KWH	R	0x0840	Calculated from 0:00 / Update every 5S (positive value for draw from grid, negative for feed to grid)
System Basic Capability Area	Total Inverter Energy (Today)	FLOAT (4 bytes)	KWH	R	0x0850	Calculated from 0:00 / Update every 5S (positive value for rectify, negative for invert)
System Basic Capability Area	Total Battery Energy (Today)	FLOAT (4 bytes)	KWH	R	0x0860	Calculated from 0:00 / Update every 5S (positive value for charge, negative for discharge)
System Basic Capability Area	Metered Grid Energy Draw (Accumulated)	FLOAT (4 bytes)	KWH	R	0x0870	Accumulated Life cycle grid-side draw energy summary
System Basic Capability Area	Metered Grid Energy Draw (Today)	FLOAT (4 bytes)	KWH	R	0x0872	Calculated from 0:00 / Update every 5S
System Basic Capability Area	Metered Grid Energy Feed (Accumulated)	FLOAT (4 bytes)	KWH	R	0x0880	Accumulated Life cycle grid-side feed energy summary
System Basic Capability Area	Metered Grid Energy Feed (Today)	FLOAT (4 bytes)	KWH	R	0x0882	Calculated from 0:00 / Update every 5S
System Basic Capability Area	Inverter AC Energy Input (Accumulated)	FLOAT (4 bytes)	KWH	R	0x0890	Accumulated inverter life cycle AC side energy input summary
System Basic Capability Area	Inverter AC Energy Input (Today)	FLOAT (4 bytes)	KWH	R	0x0892	Calculated from 0:00 / Update every 5S
System Basic Capability Area	Inverter AC Energy Output (Accumulated)	FLOAT (4 bytes)	KWH	R	0x08A0	Accumulated inverter life cycle AC side energy output summary
System Basic Capability Area	Inverter AC Energy Output (Today)	FLOAT (4 bytes)	KWH	R	0x08A2	Calculated from 0:00 / Update every 5S
System Basic Capability Area	Battery Charge Energy (Accumulated)	FLOAT (4 bytes)	KWH	R	0x08B0	Accumulated battery life cycle charged energy summary
System Basic Capability Area	Battery Charge Energy (Today)	FLOAT (4 bytes)	KWH	R	0x08B2	Calculated from 0:00 / Update every 5S
System Basic Capability Area	Battery Discharge Energy (Accumulated)	FLOAT (4 bytes)	KWH	R	0x08C0	Accumulated battery life cycle discharged energy summary
System Basic Capability Area	Battery Discharge Energy (Today)	FLOAT (4 bytes)	KWH	R	0x08C2	Calculated from 0:00 / Update every 5S
System Basic Capability Area	Captured Solar Energy (Accumulated)	FLOAT (4 bytes)	KWH	R	0x08D0	Accumulated solar panel life cycle energy captured summary
System Basic Capability Area	Captured Solar Energy (Today)	FLOAT (4 bytes)	KWH	R	0x08D2	Calculated from 0:00 / Update every 5S